CMPE 491 • SENIOR DESIGN PROJECT I • DELIVERABLE D3

MissionTwin AI – Project Backlog

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This backlog lists the planned work for CMPE 491 (Fall 2026–2027) and the main work packages for CMPE 492 (Spring 2027). Work is grouped into sprints that end at the course deadlines. Sprints 1–3 are committed; Sprints 4–7 and the CMPE 492 packages are provisional and will be refined at the planning step for each deliverable. The backlog is updated with D5 (20 November 2026) and D7 (31 December 2026), and each version is published on the project web page. Requirement identifiers (FR, NFR) and section numbers refer to the Project Specifications Report v1.3.

Column guide. Story: the item is written as a work item (No = still an epic). Sprint Ready: scope and acceptance criteria are clear enough to start. Priority: High, Medium, or Low. Story Points: initial effort estimates (about one person-hour per point), refined at sprint planning. Assigned to Sprint: committed to the sprint (Yes) or provisional (No). Sprint rows show the highest priority, the combined status, and the sum of their tasks' points. Stages (Related item): A scope and literature review; B technical simulator and MAPPO baseline; C integrated planning and fault analysis; D matched research evaluation; E final reproducible code, documentation, demonstration, and academic report.

Task Name	Related item	Story	Sprint Ready	Priority	Status	Story Points	Assigned to Sprint
Sprint 1 – Project definition and specification (28 Sep – 23 Oct 2026)	**D1, D2, D3**	**Yes**	**Yes**	**High**	**In Progress**	**120**	**Yes**
Define the project scope and research question	D1	Yes	Yes	High	**Complete**	12	Yes
Write and submit the Project Proposal	D1	Yes	Yes	High	**Complete**	24	Yes
Build the project web page	Web page	Yes	Yes	Medium	**Complete**	24	Yes
Write the Project Specifications Report with the literature review	D2	Yes	Yes	High	**Complete**	56	Yes
Publish the Specifications Report and this first backlog on the web page; submit through LMS, e-mail, and the LMS form	D2, D3	Yes	Yes	High	**In Progress**	4	Yes
Sprint 2 – Environment and baseline (26 Oct – 6 Nov 2026)	**Stage B**	**Yes**	**Yes**	**High**	**Not Started**	**66**	**Yes**
Confirm the department's standard report layout and expected report structure with the supervisor	All reports	Yes	Yes	Medium	**Not Started**	2	Yes
Confirm GPU access (university resources, EVREN, or free cloud services) with the supervisor	Section 1.2.1	Yes	Yes	High	**Not Started**	4	Yes
Set up the shared repository, Python/PyTorch environment, and pinned dependencies	NFR-03	Yes	Yes	High	**Not Started**	8	Yes

Task Name	Related item	Story	Sprint Ready	Priority	Status	Story Points	Assigned to Sprint
Reproduce a MAPPO baseline on VMAS Navigation with TorchRL or BenchMARL	FR-06, FR-07	Yes	Yes	High	Not Started	24	Yes
Run the feasibility test for local 7–14B open-weight LLMs (valid-JSON rate, latency, memory, license)	Section 3.3	Yes	Yes	High	Not Started	16	Yes
Draft the Mission Scenario and Task Assignment JSON schemas	FR-02, FR-03	Yes	Yes	Medium	Not Started	12	Yes
Sprint 3 – Analysis Report (9–20 Nov 2026)	**D4, D5**	**Yes**	**Yes**	**High**	Not Started	**92**	**Yes**
Finalize the coalition value, the mission score, and the failure protocol	FR-08, FR-09	Yes	Yes	High	Not Started	16	Yes
Define the evaluation scenarios, including heterogeneous variants	FR-02, Section 1.2.3	Yes	Yes	High	Not Started	16	Yes
Write the evaluation protocol (seeds, held-out episodes, metrics, statistics)	Section 3.3	Yes	Yes	High	Not Started	12	Yes
Extend the literature review for the analysis phase	D4	Yes	Yes	Medium	Not Started	12	Yes
Write the Project Analysis Report and update this backlog	D4, D5	Yes	Yes	High	Not Started	32	Yes
Publish D4 and D5 on the web page; submit through LMS, e-mail, and the LMS form	D4, D5	Yes	Yes	High	Not Started	4	Yes
Sprint 4 – Planning pipeline (23 Nov – 4 Dec 2026)	**Stage B**	**Yes**	**No**	**High**	Not Started	**76**	**No**
Implement the random and distance-based allocation strategies	FR-04	Yes	No	Medium	Not Started	12	No
Implement the LLM planner (prompt template, JSON output, decoding settings)	FR-01, FR-03	Yes	No	High	Not Started	24	No
Implement plan validation and the deterministic fallback with logging	FR-03, NFR-01	Yes	No	High	Not Started	24	No
Implement failure injection (zero actions, passive bodies) and task reassignment	FR-05, FR-08	Yes	No	Medium	Not Started	16	No
Sprint 5 – Execution and Shapley analysis (7–18 Dec 2026)	**Stage B**	**Yes**	**No**	**High**	Not Started	**96**	**No**
Train three-agent MAPPO policies conditioned on assigned task targets	FR-06	Yes	No	High	Not Started	32	No
Implement exact Shapley and leave-one-out analysis (eight coalitions) with unit tests	FR-09	Yes	No	High	Not Started	16	No
Implement the Shapley-informed LLM allocation strategy by passing validated contribution values to the LLM planner as planning context	FR-04, FR-09	Yes	No	High	Not Started	16	No
Implement experiment records and reproducible configurations	FR-11, NFR-03	Yes	No	Medium	Not Started	16	No
Implement credential handling and the simulation-only execution guard	NFR-04, NFR-05	Yes	No	Medium	Not Started	8	No

Task Name	Related item	Story	Sprint Ready	Priority	Status	Story Points	Assigned to Sprint
Measure plan-validation latency on designated test hardware	NFR-02	Yes	No	Low	Not Started	8	No
Sprint 6 – High-Level Design Report (21–31 Dec 2026)	**D6, D7**	**Yes**	**No**	**High**	**Not Started**	**52**	**No**
Prepare architecture diagrams (planner, validator, MAPPO, simulator interface, Shapley module)	D6	Yes	No	High	Not Started	16	No
Write the High-Level Design Report and update this backlog	D6, D7	Yes	No	High	Not Started	32	No
Publish D6 and D7 on the web page; submit through LMS, e-mail, and the LMS form	D6, D7	Yes	No	High	Not Started	4	No
Sprint 7 – Year-end exhibition (1–17 Jan 2027)	**D8**	**Yes**	**No**	**High**	**Not Started**	**48**	**No**
Integrate the prototype demonstration (simulator, MAPPO baseline, LLM planning with validation)	D8, Stage B	Yes	No	High	Not Started	24	No
Prepare and rehearse the poster and the oral presentation	D8	Yes	No	High	Not Started	24	No
CMPE 492 (Spring 2027) – provisional work packages	**Stages C–E**	**No**	**No**	**High**	**Not Started**	**TBD**	**No**
Integrated planning and fault analysis	Stage C	No	No	High	Not Started	TBD	No
Matched research evaluation across strategies and benchmarks	Stage D	No	No	High	Not Started	TBD	No
Final reproducible code, documentation, demonstration, and academic report	Stage E	No	No	High	Not Started	TBD	No
Optional extensions (MPE2 evaluation, additional fault models, dashboard)	Optional	No	No	Low	Not Started	TBD	No